

Assignment 3: Data Exploration

Bella Huang

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Exploration.

Directions

1. Rename this file `<FirstLast>_A03_DataExploration.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Assign a useful **name to each code chunk** and include ample **comments** with your code.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.
7. After Knitting, submit the completed exercise (PDF file) to the dropbox in Canvas.

TIP: If your code extends past the page when knit, tidy your code by manually inserting line breaks.

TIP: If your code fails to knit, check that no `install.packages()` or `View()` commands exist in your code.

Set up your R session

1. Load necessary packages (tidyverse, lubridate, here), check your current working directory and upload two datasets: the ECOTOX neonicotinoid dataset (`ECOTOX_Neonicotinoids_Insects_raw.csv`) and the Niwot Ridge NEON dataset for litter and woody debris (`NEON_NIWO_Litter_massdata_2018-08_raw.csv`). Name these datasets “Neonics” and “Litter”, respectively. Be sure to include the sub-command to read strings in as factors.

```
#import libraries
library(tidyverse)
library(lubridate)
library(here)
#get working directory
getwd()
```

```
## [1] "/home/guest/ENV872/EDE_Bella"
```

```
here()
```

```
## [1] "/home/guest/ENV872/EDE_Bella"
```

```
#import datasets
```

```
Neonics <- read.csv(file=here("Data/Raw/ECOTOX_Neonicotinoids_Insects_raw.csv"),stringsAsFactors = TRUE,
```

```
Litter <- read.csv(file=here("Data/Raw/NEON_NIWO_Litter_massdata_2018-08_raw.csv"),stringsAsFactors = T
```

Learn about your system

2. The neonicotinoid dataset was collected from the Environmental Protection Agency's ECOTOX Knowledgebase, a database for ecotoxicology research. Neonicotinoids are a class of insecticides used widely in agriculture. The dataset that has been pulled includes all studies published on insects. Why might we be interested in the ecotoxicology of neonicotinoids on insects? Feel free to do a brief internet search if you feel you need more background information.

Answer: Neonicotinoid is a type of pesticide, and even if it not kills insects directly, it still have a great influence in the lifecycle or reproducing patterns of insects. Since insects are very crucial to the local biodiversity and essential to the ecology, it is meaningful to study the neonicotinoid in details.

3. The Niwot Ridge litter and woody debris dataset was collected from the National Ecological Observatory Network, which collectively includes 81 aquatic and terrestrial sites across 20 ecoclimatic domains. 32 of these sites sample forest litter and woody debris, and we will focus on the Niwot Ridge long-term ecological research (LTER) station in Colorado. Why might we be interested in studying litter and woody debris that falls to the ground in forests? Feel free to do a brief internet search if you feel you need more background information.

Answer: The litter in the forest can affect the carbon storage and the woody debris are very important influencers of natural habitat. It is worth studying since they can influence the ecology.

4. How is litter and woody debris sampled as part of the NEON network? Read the NEON_Litterfall_UserGuide.pdf document to learn more. List three pieces of salient information about the sampling methods here:

Answer: 1.Trap design and placement 2.Spatial sampling design 3. Temporal sampling design

Obtain basic summaries of your data (Neonics)

5. What are the dimensions of the dataset?

```
# Number of rows and columns  
dim(Neonics)
```

```
## [1] 4623 30
```

6. Using the `summary` function on the "Effect" column, determine the most common effects that are studied. Why might these effects specifically be of interest? [Tip: The `sort()` command is useful for listing the values in order of magnitude...]

```
#summary it first
effect_summary <- summary(Neonics$Effect)

#sort
sort(effect_summary, decreasing = TRUE)
```

```
##      Population      Mortality      Behavior Feeding behavior
##      1803           1493           360           255
##      Reproduction    Development    Avoidance      Genetics
##      197             136           102           82
##      Enzyme(s)       Growth         Morphology     Immunological
##      62              38            22            16
##      Accumulation    Intoxication    Biochemistry    Cell(s)
##      12              12            11            9
##      Physiology      Histology      Hormone(s)
##      7               5              1
```

Answer: The top five most common effects are Population, Mortality, Behavior, Feeding behavior and Reproduction. These are the most important effects that are influencing the lifecycle of not only the insects but also their population developing pattern, which are closely related to the ecology of the surrounding environment.

- Using the `summary` function, determine the six most commonly studied species in the dataset (common name). What do these species have in common, and why might they be of interest over other insects? Feel free to do a brief internet search for more information if needed. [TIP: Explore the help on the `summary()` function, in particular the `maxsum` argument...]

```
#get the top 6 common studied species
summary(Neonics$Species.Common.Name, maxsum = 7)
```

```
##      Honey Bee      Parasitic Wasp Buff Tailed Bumblebee
##      667           285           183
##      Carniolan Honey Bee      Bumble Bee      Italian Honeybee
##      152           140           113
##      (Other)
##      3083
```

Answer: Among the lists, Honey Bee, Buff Tailed Bumblebee, Carniolan Honey Bee, Bumble Bee, Italian Honeybee are all bees, which are pollinators. They are the major pollinators for agricultural products and flowers, and are not pests, so it is important to see how such insecticide would affect them.

- Concentrations are always a numeric value. What is the class of `Conc.1..Author.` column in the dataset, and why is it not numeric? [Tip: Viewing the dataframe may be helpful...]

```
class(Neonics$Conc.1..Author.)
```

```
## [1] "factor"
```

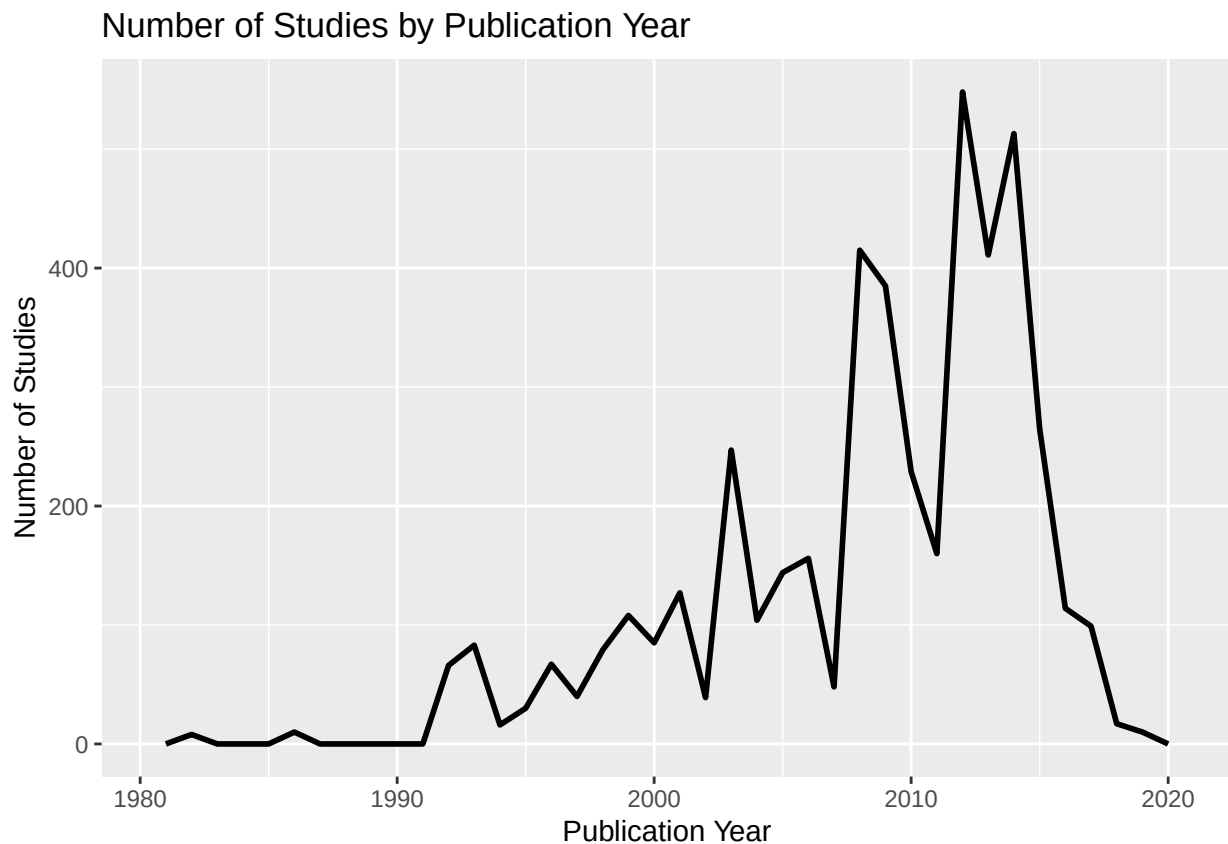
Answer: “Conc.1..Author.” is factor. When I’m looking at the dataframe, i realized that we have some value with dashes, like “144/”, which probably is the reason why prevents R from turning it into a numeric value.

Explore your data graphically (Neonics)

9. Using `geom_freqpoly`, generate a plot of the number of studies conducted by publication year.

```
library(ggplot2)
# Basic frequency polygon of publication years
ggplot(Neonics, aes(x = Publication.Year)) +
  geom_freqpoly(binwidth = 1, size = 1) +
  labs(title = "Number of Studies by Publication Year",
       x = "Publication Year",
       y = "Number of Studies")
```

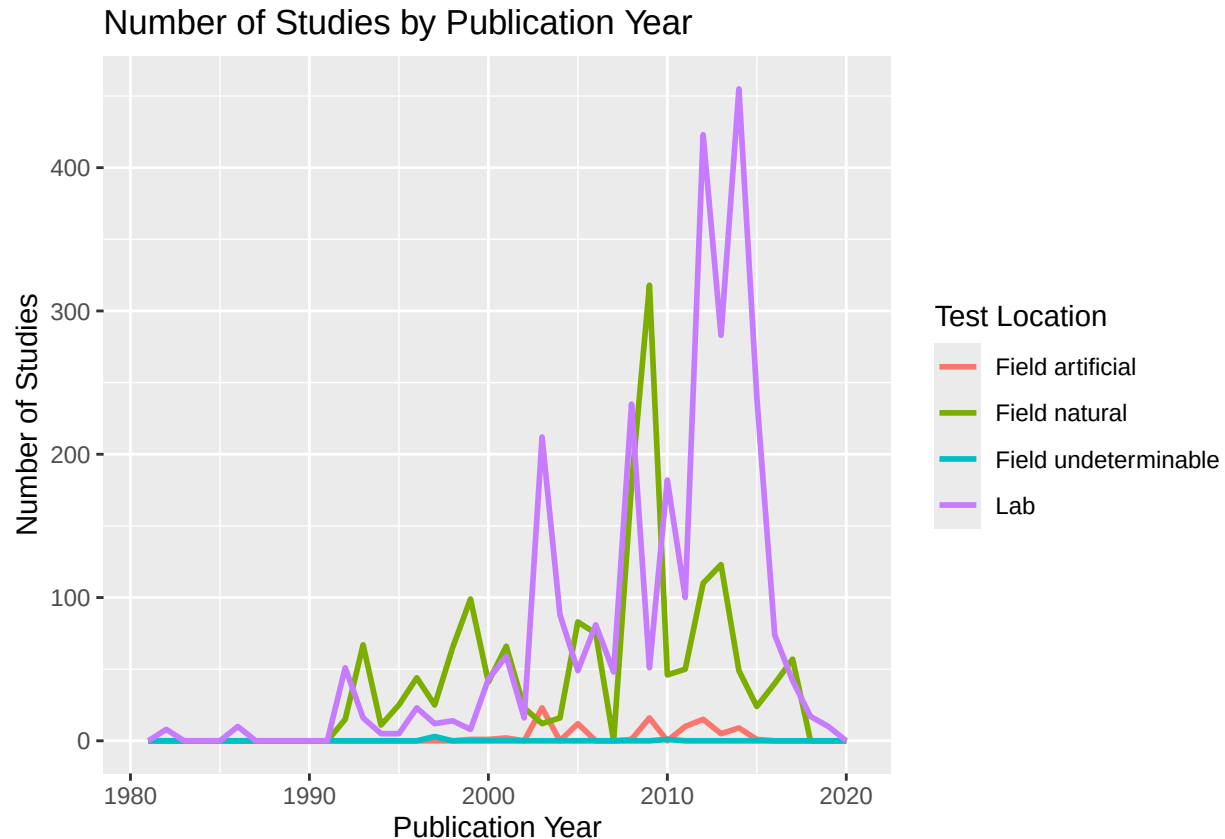
```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```



10. Reproduce the same graph but now add a color aesthetic so that different `Test.Location` are displayed as different colors.

```
#add the color aes & color legends
ggplot(Neonics, aes(x = Publication.Year, color = Test.Location)) +
  geom_freqpoly(binwidth = 1, size = 1) +
```

```
labs(title = "Number of Studies by Publication Year",
     x = "Publication Year",
     y = "Number of Studies",
     color = "Test Location")
```



Interpret this graph. What are the most common test locations, and do they differ over time?

Answer: The most common test location is Lab, especially in the period of 2000-2015. Then, the second most common one is natural field, which is pretty common in the period of 1995-2015, with its peak in around 2010. The other two artificial field and undeterminable field are rare almost all the time.

11. Create a bar graph of Endpoint counts. What are the two most common end points, and how are they defined? Consult the ECOTOX_CodeAppendix for more information.

[TIP: Add `theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))` to the end of your plot command to rotate and align the X-axis labels...]

```
library(ggplot2)
#plot the bar graph
endpoint_count <- ggplot(Neonics, aes(x = Endpoint)) +
  geom_bar() +
  labs(title = "Counts of Endpoints in Neonicotinoid Studies",
       x = "Endpoint",
       y = "# of Studies") +
```

```
theme(axis.text.x = element_text(angle = 90,
                                   vjust = 0.5,
                                   hjust = 1))
```

Answer: The two most common endpoint is LOEL and NOEL. LOEL is used for Terrestrial, and represents lowest dose (concentration) producing effects that were significantly different from responses of controls according to the appendix. NOEL is also used for Terrestrial, on the other hand, it represents highest dose (concentration) producing effects not significantly different from responses of controls according to author's reported statistical test.

Explore your data (Litter)

12. Determine the class of collectDate. Is it a date? If not, change to a date and confirm the new class of the variable. Using the `unique` function, determine which dates litter was sampled in August 2018.

```
#check the class --> factor
class(Litter$collectDate)
```

```
## [1] "factor"
```

```
#change to a date
Litter$collectDate <- as.Date(as.character(Litter$collectDate),
                              format = "%Y-%m-%d")
```

```
#check the unique value
unique(Litter$collectDate)
```

```
## [1] "2018-08-02" "2018-08-30"
```

13. Using the `unique` function, list the different plotIDs sampled at Niwot Ridge. How is the information obtained from `unique` different from that obtained from `summary`?

```
#unique
unique(Litter$plotID)
```

```
## [1] NIWO_061 NIWO_064 NIWO_067 NIWO_040 NIWO_041 NIWO_063 NIWO_047 NIWO_051
## [9] NIWO_058 NIWO_046 NIWO_062 NIWO_057
## 12 Levels: NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 ... NIWO_067
```

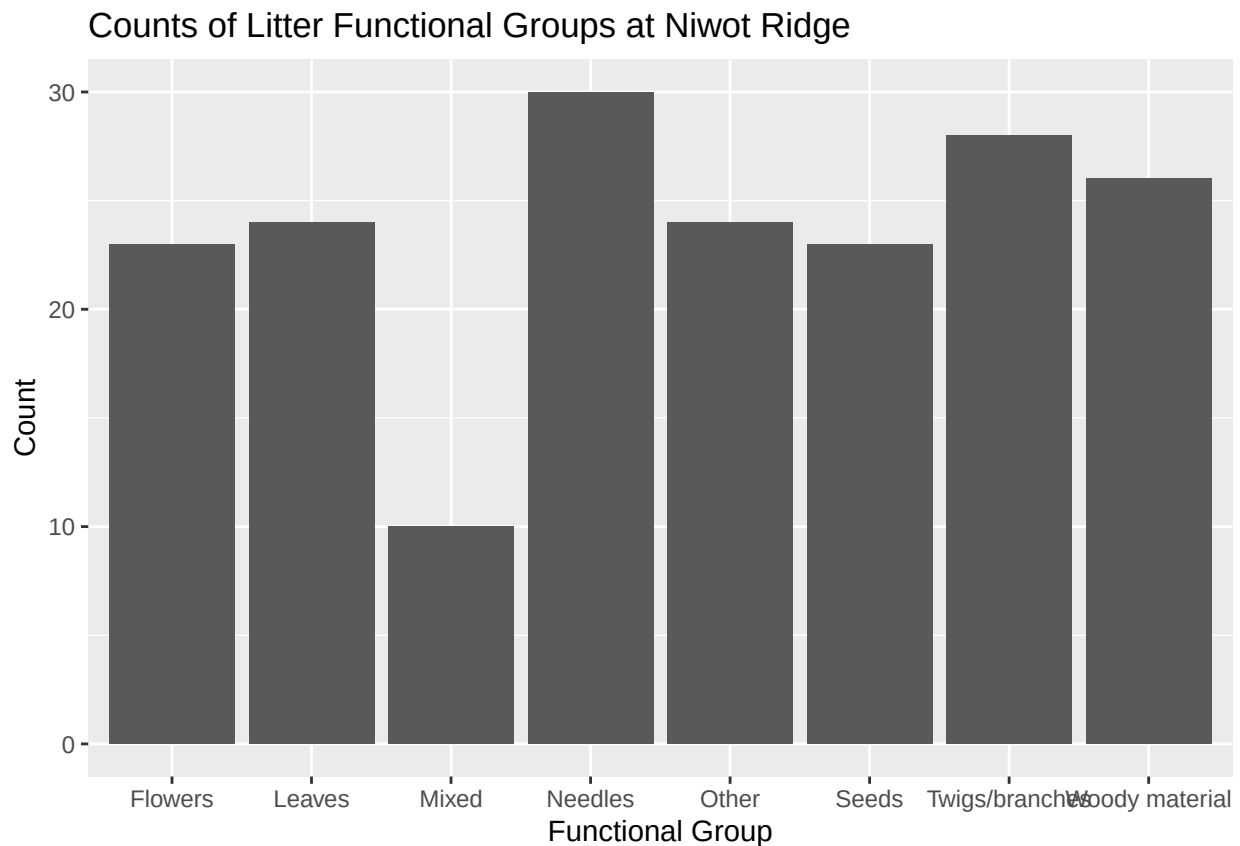
```
#summary
summary(Litter$plotID)
```

```
## NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 NIWO_058 NIWO_061
##      20      19      18      15      14      8      16      17
## NIWO_062 NIWO_063 NIWO_064 NIWO_067
##      14      14      16      17
```

Answer: Unique only gives me the values that exists in the column, while Summary shows how they are distributed, for instance the count of values.

14. Create a bar graph of functionalGroup counts. This shows you what type of litter is collected at the Niwot Ridge sites. Notice that litter types are fairly equally distributed across the Niwot Ridge sites.

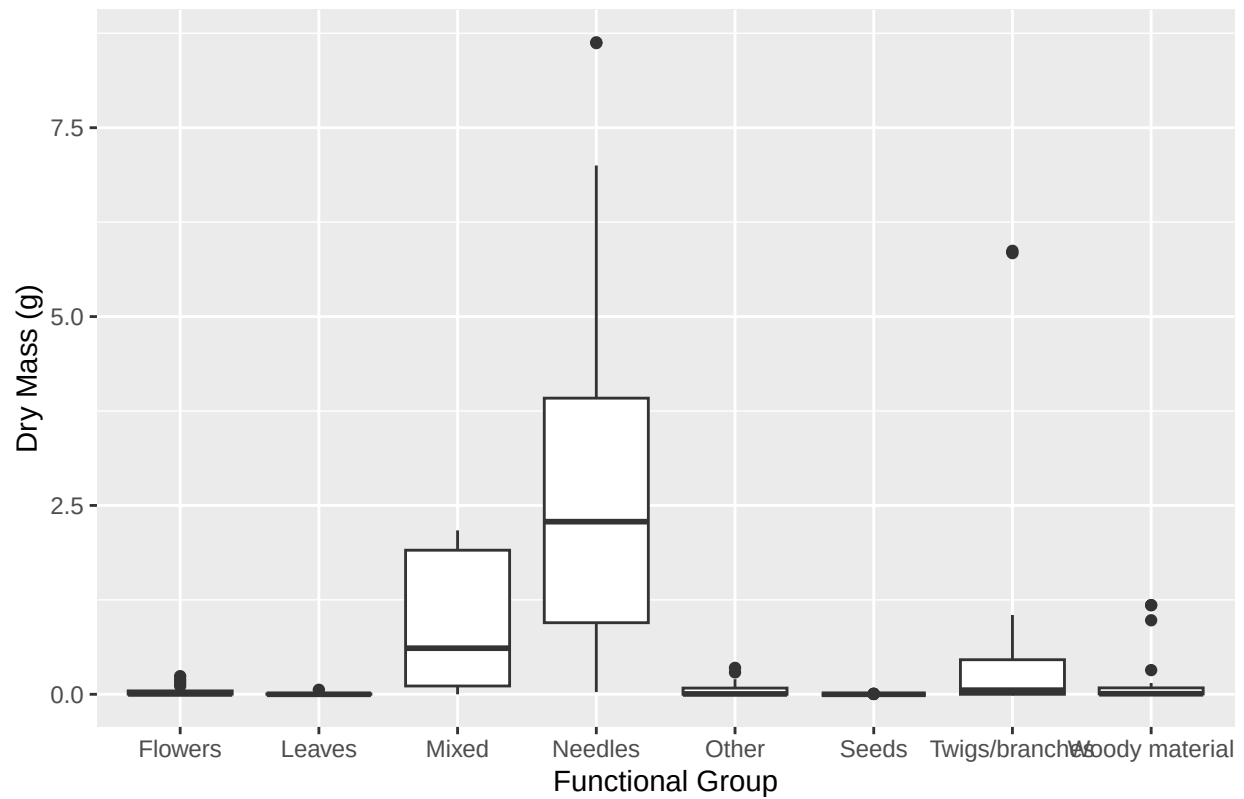
```
ggplot(Litter, aes(x = functionalGroup)) +  
  geom_bar() +  
  labs(title = "Counts of Litter Functional Groups at Niwot Ridge",  
        x = "Functional Group",  
        y = "Count")
```



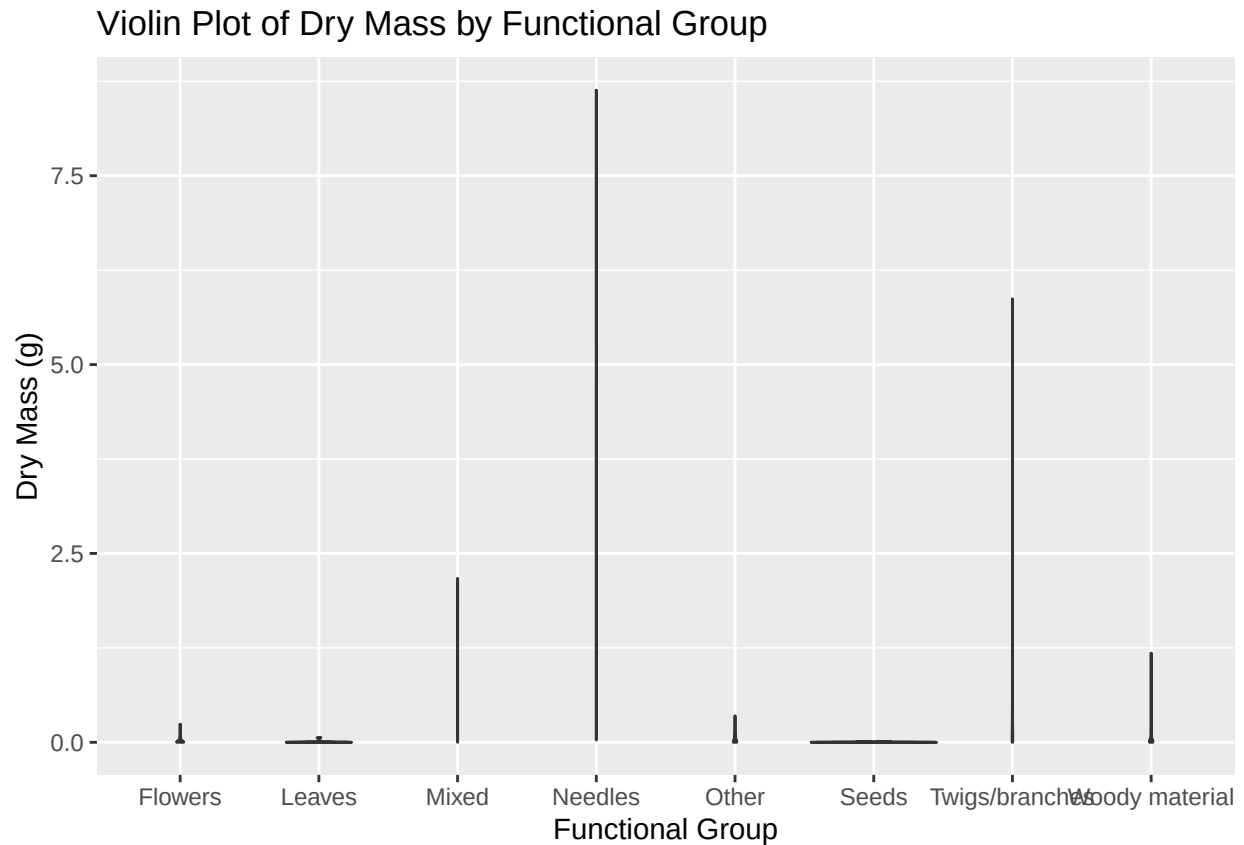
15. Using `geom_boxplot` and `geom_violin`, create a boxplot and a violin plot of `dryMass` by `functionalGroup`.

```
#Box plot  
ggplot(Litter, aes(x = functionalGroup, y = dryMass)) +  
  geom_boxplot() +  
  labs(title = "Boxplot of Dry Mass by Functional Group",  
        x = "Functional Group",  
        y = "Dry Mass (g)")
```

Boxplot of Dry Mass by Functional Group



```
#Violin plot
ggplot(Litter, aes(x = functionalGroup, y = dryMass)) +
  geom_violin(draw_quantiles = c(0.25, 0.5, 0.75)) +
  labs(title = "Violin Plot of Dry Mass by Functional Group",
       x = "Functional Group",
       y = "Dry Mass (g)")
```



Why is the boxplot a more effective visualization option than the violin plot in this case?

Answer: Since we do not have a very large sample size, using violin plot rather than boxplot can overcomplicate the situation. The boxplot just gives us straight forward information and tendency, which are very clear to see.

What type(s) of litter tend to have the highest biomass at these sites?

Answer: Needles and Seeds Twigs.